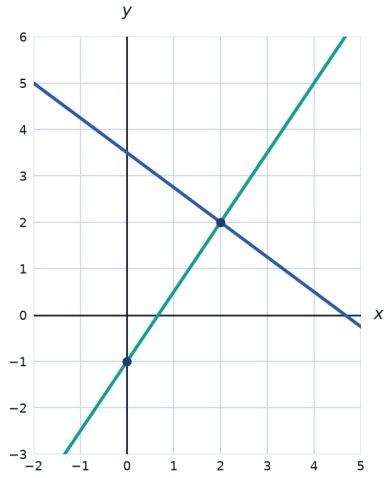
# Graph Theory & Networks (AI HL)

## Section 1: Degrees and the Handshaking Lemma

**1** What does the Handshaking Lemma state for any undirected graph with $V$ vertices and $E$ edges?




**a.** The sum of the degrees of all vertices equals twice the number of edges: $\sum \deg(v) = 2E$

**b.** The sum of the degrees equals $E$

**c.** The number of vertices with odd degree is always odd

**d.** Every vertex must have degree at least 2

**2** A graph has $6$ vertices, each having degree $3$. How many edges does the graph have?

**a.** $9$

**b.** $18$

**c.** $6$

**d.** $12$

**3** Why can a graph never have an odd number of vertices of odd degree?

**a.** Because $\sum \deg(v) = 2E$ is always even, the sum of odd degrees must be even, forcing an even count of odd vertices

**b.** Because edges cannot cross each other

**c.** By Euler's formula $V - E + F = 2$

**d.** Because every vertex must have at least one adjacent edge

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## Section 2: Eulerian and Hamiltonian Paths and Circuits

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**4** What condition is both necessary and sufficient for a connected graph to contain an Eulerian circuit (a closed trail visiting every edge exactly once)?

**a.** Every vertex has an even degree

**b.** Exactly two vertices have odd degree

**c.** The graph contains no cycles

**d.** The graph is a complete bipartite graph

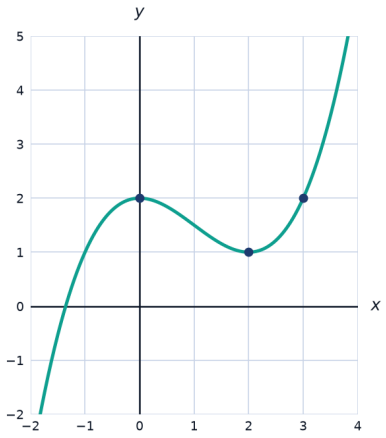
**5** If a connected graph has exactly two vertices of odd degree, what type of Eulerian walk does it contain?

**a.** An Eulerian trail (open path visiting every edge once) that starts at one odd vertex and ends at the other

**b.** An Eulerian circuit returning to the start

**c.** No Eulerian trail or circuit can exist

**d.** A Hamiltonian cycle




**6** What is the difference between an Eulerian circuit and a Hamiltonian cycle?

**a.** An Eulerian circuit visits every edge once; a Hamiltonian cycle visits every vertex once

**b.** An Eulerian circuit visits every vertex once; a Hamiltonian cycle visits every edge once

**c.** They are identical concepts in graph theory

**d.** Hamiltonian cycles apply only to directed trees

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## Section 3: Minimum Spanning Trees (Kruskal & Prim)

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**7** In a connected weighted graph with $V$ vertices, how many edges does any spanning tree contain?

**a.** $V - 1$

**b.** $V$

**c.** $V + 1$

**d.** $\frac{V(V-1)}{2}$

**8** How does Kruskal's algorithm select edges to construct a Minimum Spanning Tree (MST)?

**a.** Sorts all edges by weight and greedily adds the smallest available edge that does not create a cycle

**b.** Grows a single tree by continuously adding the closest vertex to the current tree

**c.** Removes the largest cycle edges one by one until only a tree remains

**d.** Visits vertices in alphabetical order

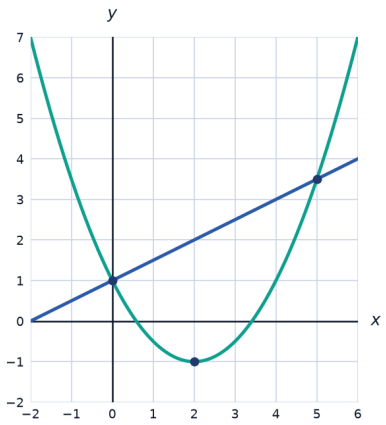
**9** How does Prim's algorithm differ conceptually from Kruskal's algorithm?

**a.** Prim grows a single connected tree from an arbitrary starting vertex, whereas Kruskal maintains a forest of components until unified

**b.** Prim works only on unweighted graphs

**c.** Kruskal finds the longest path rather than minimum tree

**d.** Prim can only be applied to planar graphs




## Section 4: Shortest Paths and Dijkstra's Algorithm

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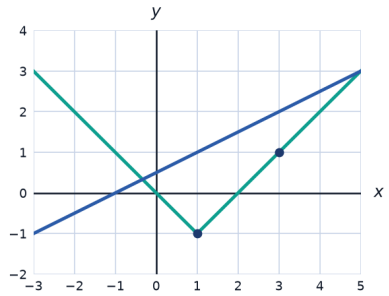
**10** What is the purpose of Dijkstra's algorithm?

**a.** To find the shortest paths from a single source vertex to all other vertices in a network with non-negative edge weights

**b.** To find a minimum spanning tree

**c.** To determine if a graph has an Eulerian circuit

**d.** To find the maximum flow through a network




**11** Why does Dijkstra's algorithm fail if a network contains negative edge weights?

**a.** A greedy choice once finalized could be improved later by a negative edge, invalidating the optimality assumption

**b.** Distances cannot be negative by definition

**c.** The algorithm enters an infinite cycle immediately

**d.** Square roots of negative weights cannot be computed

**12** In a planar graph, if $V = 6$ and $E = 9$, how many faces $F$ does the graph enclose according to Euler's formula $V - E + F = 2$?

**a.** 5

**b.** 3

**c.** 4

**d.** 6

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